

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
Summer Semester September 2023
22MCA103 – COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each Unit

Unit - I		Marks	BT*	CO*	PO*
1. a)	What is binary-coded decimal and how is it used?	6	2	1	1
b)	Prove Demorgan's theorem using truth table.	10	2	1	1
C)	Simplify the Boolean expression $A'(A + B) + (B + AA)(A + B')$	4	2	1	1
2. a)	Prove the following using Boolean theorem	6			
	i) $X+XY = X$				
	ii) $X.0 = 0$				
	iii) $X+X = X$		3	L	2
b)	With truth table explain three input XOR and XNOR gate and implement XNOR gate using only NAND gates.	8	3	L	2
C)	Express the Boolean function $F=A+BC$ as minterm. Indicate all the steps	6	2	L	1
UNIT-II					
3. a)	Simplify the following expression using Boolean algebra $(A + B' + C')(A + B' + C)(A + B + C')$	6	3	2	2
b)	Simplify the following expression using K-Map $F(A,B,C,D) = A'B'C'D' + A'B'CD' + A'BCD' + A'BCD + AB'C'D' + ABCD'$	7	3	2	2
c)	Identify the number of prime implicants and essential prime implicants for the function. $f(A, B, C, D) = \sum m(0, 1, 4, 6, 7, 8, 10, 14, 15)$. Using tabulation method.	7	3	2	2
4. a)	Implement the following expression using 8:1 mux $F(A,B,C,D)=\sum m(0,1,3,5,7,10,11,13,14,15)$. using	10	3	2	2
b)	Implement full adder with a 3:8 decoder	10	3	2	2
UNIT-III					
5. a)	Perform following operations using 2's complement arithmetic				
	i) $(7)_{10} - (15)_{10}$	10	3	3	2
	ii) $(50)_{10} - (2A)_{16}$	10	3	3	2
b)	Design 4-bit 2's complement Adder/Subtractor circuit.	10	3	3	2
6. a)	Explain restoring division algorithm & use it to divide 1000/11	10	3	3	2
b)	With a neat circuit diagram, explain the working of half subtractor and Full Subtractor.	10	2	3	1
UNIT-IV					
7. a)	Write assembly language program to multiply two 16 bit numbers stored in memory	10	3	4	2
b)	Explain the push and pop operation of stack with neat diagram and an example	10	2	4	1
8. a)	What are addressing modes? Explain any 5 types of addressing modes.	10	2	4	1
b)	Show how $X = (A+B)*(C+D)$ can be implemented using two and three address instructions.	10	3	4	2
UNIT-V					
9. a)	Define Bus arbitration. Explain the two approaches to bus arbitration.	10	2	5	1
b)	Analyze how data are written into ROM. Discuss different types of ROM.	10	2	5	1
10. a)	Explain internal structure of synchronous DRAM.	10	2	5	1
b)	What is cache memory. Analyze three mapping functions of cache memory.	10	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
